

APPLIED AI & ROBOTICS · HOUSTON CITY COLLEGE

Michelle Adu-Asah

Building Intelligent Systems From NLP Pipelines to Neural Networks

github.com/michelleaa68-ux

From Foundations to NLP

Six courses building toward a complete understanding of intelligent systems from AI philosophy to deep learning and natural language processing.



NewsBot 2.0

An NLP pipeline that evolved from a midterm prototype into a full-featured, multilingual final project.

NewsBot NLP Midterm

Text classification with Naive Bayes and TF-IDF. Sentiment analysis, POS tagging, syntax parsing, and named entity recognition.

- 96.3% accuracy on BBC News dataset
- NLTK, spaCy, scikit-learn, pandas

NewsBot 2.0 NLP Final

Upgraded end-to-end NLP pipeline. Adds LDA topic modeling, LSA summarization, semantic search, multilingual translation, and a conversational chatbot interface.

- 96%+ accuracy · 1,490 articles · 4 languages
- NLTK, spaCy, scikit-learn, sumy, langdetect, deep-translator

96%+ Accuracy

4 Languages

English · German · French · Spanish

[Project Repository](#)

[Live Demo](#)

MIRA: Multi-Agent Event Decor Intelligence System

A rule-based multi-agent AI system built for a luxury event decor company, routing clients and workers to three specialized agents.

Agent Roles

- MIRA Client Q&A, pricing & booking
- MIZEL Creative vision & mood boards
- MILO Worker inventory (passcode protected)

Technical Build

- Rule-based logic with keyword matching
- In-session memory for inventory tracking
- Role-based routing system
- Built in Python / Google Colab

3 Agents

MIRA · MIZEL · MILO

1 System

Client · Creative · Inventory

[Project Repository](#)

[Live Demo](#)

Technical Skills

- 1Python
- 2NLTK
- 3spaCy
- 4scikit-learn
- 5PyTorch
- 6Hugging Face
- 7VS Code
- 8Google Colab
- 9HTML/CSS
- 10CNNs
- 11RNNs
- 12Transformers
- 13Multi-agent Systems
- 14NLP Pipelines
- 15AI Cybersecurity
- 16Database Design

What Makes Me Unique



Multilingual by Nature
Born and raised in Germany, fluent in multiple languages, built multilingual AI because I understand what language means personally.



Design Meets AI
Trained in graphic design and 3D design at a vocational college in Germany I build systems that work and look professional.



Security Aware
Studied AI Cybersecurity I understand the risks behind the systems I build, not just how to build them.

Looking Back, Moving Forward

What I Learned

- How to build real AI systems from scratch
- The full ML workflow end-to-end
- How chatbots, translators, and agents work under the hood
- The ethics and security side of AI

Where I'm Headed

- Build a web app for NewsBot 2.0
- Explore BERT and transformer models deeper
- Apply AI skills professionally
- Keep growing this portfolio